# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
## Department of Computer Science and Engineering
## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

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|---|---|---|---|
| **Course Code:** | CSA06 | **Course Title:** | Design and Analysis of Algorithms |
| **CO Assessed:** | CO2 – Manipulation (BL1, BL2, BL3) | **Assessment:** | Assessment Tool 1 – Analytical Problem Solving |
| **Weightage:** | 40% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)* | | |
| **Student Name:** | Sasikumar Megha | **Reg. No.:** | 192471038 |
| **Section / Batch:** | CS & BS | **Date:** | 3/8/26 |

**Instructions to Students**
- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

| Question Type | Focus Area | Questions |
|---|---|---|
| **Application-Based Questions** | Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis | Q1 |

## SECTION A – APPLICATION BASED QUESTIONS

90/100

**Code of Conduct**
*I Sasikumar Megha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S. Megha

## RUBRICS FOR CALCULATION

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20% | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30% | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20% | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20% | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10% | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

## Marks Evaluation:

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20% | ✓ | | | |
| Application of Concepts | 30% | ✓ | | | |
| Problem-Solving Approach | 20% | | ✓ | | |
| Accuracy of Solution | 20% | ✓ | | | |
| Clarity | 10% | | ✓ | | |
| **Total Marks (100):** | | 90 | | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| Name: Rizwan [illegible] | Name: | Name: |
| Signature: | Signature: | Signature: |
| Date: 3/8/26 | Date: | Date: |

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CSA0601

# Co2- AT-1

Q1) Selection Sort - Dataset with scattered small element.

Dataset : [43, 9, 37, 14, 28, 5]

Passes

* Pass 1: [5, 9, 37, 14, 28, 43]
* Pass 2: [5, 9, 37, 14, 28, 43]
* Pass 3: [5, 9, 14, 37, 28, 43]
* Pass 4: [5, 9, 14, 28, 37, 43]
* Pass 5: [5, 9, 14, 28, 37, 43]

Sorted Array: [5, 9, 14, 28, 37, 43]

* Comparisons : 15
* Swaps : 3
* Time complexity : Best = Average = worst = O(n

92) Bubble Sort - Progressive Exchange Analysis

Dataset : [34, 12, 29, 45, 18, 7]

Passes :

* Pass 1 : [12, 29, 34, 18, 7, 45]
* Pass 2 : [12, 29, 18, 7, 34, 45]
* Pass 3 : [12, 18, 7, 29, 34, 45]
* Pass 4 : [12, 7, 18, 29, 34, 45]
* Pass 5 : [7, 12, 18, 29, 34, 45]

Sorted array : [7, 12, 18, 29, 34, 45]

* Comparisons : 15
* Swaps : 10
* Time complexity : Best = $O(n)$, Average = $O(n^2)$, worst = $O(n^2)$

93) Linear Seach :

Dataset : [22, 31, 47, 58, 16, 39, 64]

Search for 39

22 X

31 X

47 X

58 X

16 ×

39 ✓

Result: found at index 5

* comparison: 6

* Time complexity: Best = O(1), Average = O(n), worst - O(n)

Q.94) Binary Search

Dataset: [5, 10, 15, 20, 25, 30, 35, 40]

Search for 25

* Mid = 20 → Right half

* Mid = 30 → Left half

* Mid = 25 → found

Result: found at index 4

* comparison: 3

* Time complexity: Best = O(1), Average = O(log n) worst = O(log n)

Closest pair

points : (1,3), (4,5), (7,8), (2,4), (9,1), (5,6)

using distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

closest pairs :

* (1,3) and (2,4) → Distance = 1.41
* (4,5) and (5,6) → Distance = 1.41
* Total comparison : 15
* Time complexity : $O(n^2)$